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To Reason but Not to Stop: Understanding and Recovering Pruned Large Reasoning Models

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Abstract

Weight pruning can degrade the accuracy of large reasoning models (LRMs), yet it remains unclear whether it damages reasoning itself. We find that a substantial fraction of this degradation stems from a failure to *stop* rather than to *reason*. As sparsity increases, pruned LRMs increasingly fail to terminate within the token budget, often after the correct answer has appeared in their reasoning, instead falling into repetitive loops, even though they still recognize where the dense model would stop. We further find that the hidden-state difference between terminating and looping continuations is consistent across pruned checkpoints, suggesting a shared direction associated with committing to an answer. Building on this, we propose Termination-Guided Activation Steering (TGAS), a training-free method that adds this direction to the residual stream at inference time. Across six LRMs, three pruning pipelines, and math and code benchmarks, TGAS improves termination and overall accuracy while reducing repetitive reasoning and generation length, showing that this failure is actionable.

1 Introduction

Large reasoning models (LRMs) achieve strong performance by generating long chain-of-thought traces, but this makes inference expensive (Zhang et al., 2026a; Guo et al., 2025). Weight pruning offers a natural approach to reduce cost by zeroing out selected parameters (Frantar and Alistarh, 2023; Sun et al., 2024). However, as consistently observed across pruning methods, pruning LRMs can substantially degrade accuracy on reasoning tasks (Ji et al., 2025; Lucas et al., 2026; Zhang et al., 2026b). Existing efforts have primarily focused on mitigating this degradation through improved pruning calibration (Ji et al., 2025; Lucas et al., 2026; Zhang et al., 2026b), alongside post-pruning recovery via retraining or fine-tuning (Liu et al., 2025b; Muralidharan et al., 2024; Kim et al., 2024; Xia et al., 2024a). Yet, it remains unclear whether this degradation reflects a loss of reasoning capability or a different kind of failure.

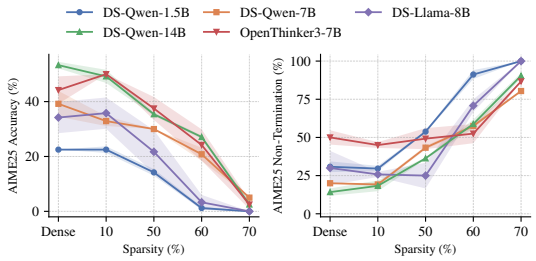


Figure 1: Accuracy (left) and non-termination rate (right) on AIME-25 across pruning settings (SparseGPT with OBC calibration, 16,384-token budget, averaged over 4 runs). Accuracy decreases as non-termination becomes more frequent.

We find that much of this degradation stems from a failure to *stop* rather than a failure to *reason*. As sparsity increases, pruned LRMs increasingly fail to terminate and commit to a final answer within the token budget, and accuracy falls accordingly (Figure 1). Strikingly, many of these non-terminating traces already contain the correct answer, yet the model keeps reasoning until the budget is exhausted. We refer to this behavior as ***reasoning divergence***. On problems solved by the dense model, pruned models remain largely accurate whenever they terminate, and non-termination becomes the dominant source of errors as sparsity grows (Section 3.1).

Why do pruned models fail to stop? Pruning does not erase the stopping signal: when replaying dense traces, pruned models still assign high probability to the end-of-thinking token where the dense model stops. Instead, during free-running generation they increasingly revisit earlier reasoning rather than moving forward, and fall into repetitive loops. Moreover, the hidden-state difference between continuations that terminate and

those that loop points in a similar direction across pruned checkpoints, suggesting that the shift between looping and committing to an answer is captured by a shared direction in representation space.

Based on this diagnosis, we propose *Termination-Guided Activation Steering* (TGAS), which adds this direction to the residual stream at inference time to steer pruned models toward termination without retraining or weight updates. TGAS consistently improves termination and accuracy across six LRMs, three pruning pipelines, and math and code benchmarks by reducing repetitive reasoning. Our contributions are threefold: (i) We show that a substantial part of pruning-induced degradation in LRMs arises from reasoning divergence, a failure to commit rather than a loss of reasoning ability; (ii) We identify a hidden-state direction that separates terminating from looping continuations, and build on it TGAS, a training-free steering method; (iii) We show that TGAS consistently improves termination and overall accuracy across models, pruning methods, and benchmarks by reducing looping.

2 Background and Related Work

Pruning for LRMs. One-shot unstructured pruning formulates compression as layer-wise reconstruction (Frantar et al., 2022): given a weight matrix $\mathbf{W}_\ell$ and calibration inputs $\mathbf{X}_\ell$, it solves $\min_{\|\hat{\mathbf{W}}_\ell\|_0 \le S} \|(\mathbf{W}_\ell - \hat{\mathbf{W}}_\ell)\mathbf{X}_\ell\|_F^2$, where S is the number of retained weights. We focus on two retraining-free methods: SparseGPT (Frantar and Alistarh, 2023), which uses second-order information to compensate for pruning errors, and Wanda (Sun et al., 2024), which scores weights by magnitude and input-activation norm. To better capture long reasoning traces, recent LRM pruning methods improve calibration via on-policy chain-of-thought RAC (Lucas et al., 2026), reasoning-step importance OBC (Nguyen et al., 2026), or self-generated traces (Ji et al., 2025; Wang et al., 2025; Xiang et al., 2025), detailed preliminaries in Appendix B.

Related Works. LRMs can overthink and enter repetitive loops (Chen et al., 2026; Pipis et al., 2026), which quantization can further prolong (Lotfi et al., 2026). We show that pruning amplifies looping while largely preserving the stopping signal, distinguishing failures to stop from failures to reason. Activation steering offers a way to modify behavior by adding directions to the residual stream, with difference-in-means directions shown to control behaviors of LLMs (Arditi et al., 2024). For reasoning, SEAL (Chen et al., 2025) suppresses reflection, while Plan-CoT (Sheng et al., 2025) modulates reasoning length. In contrast, TGAS targets pruning-induced non-termination using contrasting continuations of the same prefix, yielding a direction weakly aligned with SEAL's (Appendix F.8).

3 Understanding Pruning-Induced Reasoning Degradation

We investigate three questions: (i) Do pruned models lose reasoning ability or only the ability to stop? (ii) Is non-termination caused by limited generation budgets? (iii) Why do models fail to stop despite reaching the correct answer? We address the first two questions in Section 3.1 and the third in Section 3.2. Recall that we refer to this phenomenon as *reasoning divergence*, characterized by *non-termination*, driven by repetitive *looping* and an *failure to commit* to final answers.

Setting. We evaluate DeepSeek-R1-Distill-Qwen (DS-Qwen), DeepSeek-R1-Distill-Llama (DS-Llama) (Guo et al., 2025), and OpenThinker3 (OT3) (Guha et al., 2026) under one-shot unstructured pruning of decoder linear layers at varying sparsity levels, using SparseGPT or Wanda with OBC, RAC, or C4 calibration. Benchmarks include MATH-500 Levels 4-5 (hereafter MATH-500) (Lightman et al., 2024), AIME-25 (Zhang and Math-AI, 2025), and LiveCodeBench-v6 (LCB) (Jain et al., 2025); OpenR1-Math-220K (OpenR1-Math) (Hugging Face, 2025) is used only for calibration and steering-direction construction. We use a 16,384-token budget, $T = 0.6$, and top-$p = 0.95$, averaging $k = 4$ responses on three benchmarks. Our default uses SparseGPT with OBC calibration on model-generated reasoning traces, counting a response as correct only if it commits to a final answer, detailed in Appendix C.

3.1 Do Pruned Models Lose the Ability to Reason?

Accuracy on Dense-Correct Problems. To determine whether reasoning divergence reflects a loss of reasoning capability, we evaluate pruned models on problems correctly solved by their

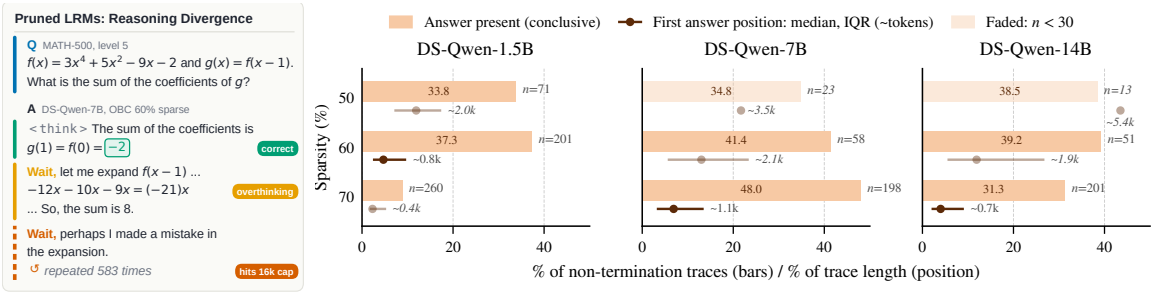


Figure 2: Gold answers in non-terminating reasoning traces of DeepSeek models on MATH-500. Left: an example of reasoning divergence, where the pruned model continues reasoning after reaching the correct answer. Right: the fraction of traces containing the gold answer and the median position of its first occurrence. n denotes the number of non-terminating samples.

Model	Ckpt	Kept	No-Term.	Wrong	Kept|term
DS-Qwen -7B	obc_50	91.7±0.8	3.8±0.3	4.5±0.6	95.3±0.6
	obc_60	77.8±0.7	16.8±0.1	5.4±0.6	93.5±0.7
DS-Qwen -14B	obc_50	89.2±0.6	2.4±1.5	8.4±0.9	91.4±0.8
	obc_60	74.3±2.8	16.4±1.9	9.3±0.9	88.9±1.4
OT3-7B	obc_50	79.3±2.9	13.1±2.4	7.6±1.9	91.3±2.2
	obc_60	63.6±3.4	29.8±3.3	6.6±1.8	90.6±2.5

Table 1: Accuracy on MATH-500 problems correctly solved by the dense model. **Kept**: correct; **No-Term.**: reasoning fails to terminate; **Wrong**: terminated but incorrect; **Kept|term**: accuracy among terminated generations. From 50% to 60% sparsity, non-termination grows sharply while the terminated-but-wrong rate stays nearly flat, so accuracy among terminated generations remains high.

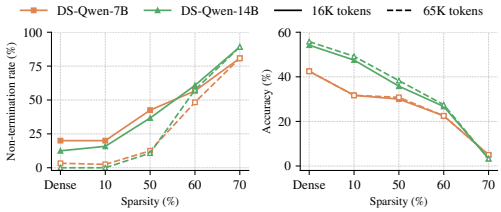


Figure 3: Non-termination rate and accuracy of DeepSeek models on AIME-25 under 16K- and 65K-token budgets.

dense counterparts. Table 1 shows that as sparsity grows, the additional errors come mainly from non-termination rather than from incorrect answers. On MATH-500, going from 50% to 60% sparsity raises the non-termination rate from 3.8% to 16.8% for DS-Qwen-7B, and from 2.4% to 16.4% for the 14B counterpart, while the terminated-but-wrong rate stays nearly unchanged. Terminated generations largely preserve their accuracy at high sparsity, despite fewer dense-correct samples remaining correct. These results suggest that substantial reasoning capability remains intact under moderate pruning, while termination becomes the primary bottleneck as sparsity grows (full results in Appendix D.1).

Larger Generation Budget. We ask whether reasoning divergence is simply caused by an insufficient generation budget. As shown in Figure 3, increasing the budget from 16,384 to 65,536 tokens substantially reduces non-termination at low and moderate sparsity, e.g., from 42.5% to 12.5% for 7B at 50% sparsity, but leaves accuracy nearly unchanged for DS-Qwen 7B and 14B, while at 60–70% sparsity non-termination barely decreases. Appendix D.2 shows that the larger budget recovers only 0–3 additional correct traces across settings, suggesting that pruning-induced failures are not primarily due to insufficient generation length.

Correct Answer Presence in Reasoning Traces. Since a larger budget rarely helps, we next ask what the non-terminating traces contain: does the reasoning already reach the correct answer even when the model fails to terminate and finalize it? To answer this question, we use text-based rules, termed *Conclusive*, that detect answer-like statements following cues such as *"thus the answer"* or *"so we get"*(Appendix C.2).[1] Figure 2 shows that a substantial fraction of non-terminating traces already contain the correct answer: at 60% sparsity, this holds for 41.4% and 39.2% of non-terminating traces for DS-Qwen 7B and 14B, respectively, with the answer appearing early at a median of 13.0% and 11.9% of the trace length, roughly 2.1k and 1.9k tokens. Despite reaching the correct answer, models continue generating to the token limit, often entering repetitive or corrective reasoning. This suggests that non-termination often reflects reasoning divergence rather than an inability to reach the correct answer. We provide a headroom analysis in Appendix E.3, quantifying the recoverable accuracy loss by counting non-terminating traces that already contain the correct

[1] We validate *Conclusive* against an independent LLM judge: on 250 sampled traces, the two agree on 96.0% of cases ($\kappa = 0.898$; Appendix C.3).

		DS-Qwen-7B			DS-Qwen-14B		
		obc_50	obc_60	obc_70	obc_50	obc_60	obc_70
AIME-25	p(</think>)	0.998	0.957	0.036	0.995	0.854	0.073
	Med. rank	0	0	5.5	0	0	2
MATH-500	p(</think>)	0.996	0.924	0.040	0.998	0.907	0.130
	Med. rank	0	0	7	0	0	3

Table 2: Teacher-forced stopping signal at the dense model's STOP position: probability of </think> and its median rank (0 means </think> is the top prediction).

	DS-Qwen-7B				OT3-7B			
	dense	obc_50	obc_60	obc_70	dense	obc_50	obc_60	obc_70
n	794	788	796	782	800	800	796	790
Δ_{cyc}	−15.3	−11.9	−9.6	−5.0	−23.1	−15.8	−13.2	−5.6
Entropy	1.55	1.51	1.41	1.44	1.03	1.22	1.33	1.60

Table 3: Log-probability ratio between cyclic and progressive continuations at decision points on AIME-25. n: number of decision points; **Entropy**: predictive entropy at the decision points. Less negative (larger) Δ_{cyc} values indicate a stronger preference for the repetitive continuation.

answer as correct.

3.2 Why Do Pruned Models Fail to Terminate?

Section 3.1 shows that pruned LRMs largely retain the ability to reach correct answers but increasingly fail to terminate. We show that the stopping signal remains intact, yet during free-running generation, models struggle to commit to final answers and fall into repetitive reasoning loops.

Does pruning damage the stopping mechanism? We test whether pruning impairs stopping recognition via teacher forcing. Let $\mathcal{D}_{\text{cov}}$ denote dense traces that terminate correctly. We replay these traces through each pruned checkpoint and measure p(</think>) and its rank at the dense model's termination position (**STOP**). As shown in Table 2, at 50–60% sparsity, the signal remains strong: for 7B, p(</think>) $>$ 0.92 with median rank 0 on both AIME-25 and MATH-500, while 14B exceeds 0.85 with the same rank. At 70%, it drops to 0.036–0.040 for 7B and 0.073–0.130 for 14B. Comparing STOP with positions more than 50 tokens earlier, we find that </think> remains substantially more likely at STOP at all sparsities, but at 70% its log-probability falls below that of the ordinarily generated tokens, detailed in Appendix D.3. Thus, the stopping signal remains largely intact under moderate pruning, suggesting that the main failure is reaching, rather than recognizing, the stopping point. At 70% sparsity, the signal remains localized but weakens, consistent with severe pruning also impairing stopping recognition.

Definition. We investigate whether pruning alters reasoning trajectories at moments of reconsideration, where a model may either move forward or revisit previous reasoning. We define a *decision point* as the start of a paragraph beginning with a predefined reconsideration cue (e.g., *"wait"*, *"hmm"*, full list in Appendix D.4), following Lotfi et al. (2026). We label each paragraph CYCLIC if its 8-word n-gram set has Jaccard similarity ≥ 0.5 with any earlier paragraph in the same trace, and PROGRESS otherwise, following Pipis et al. (2026), as illustrated in Figure 4.

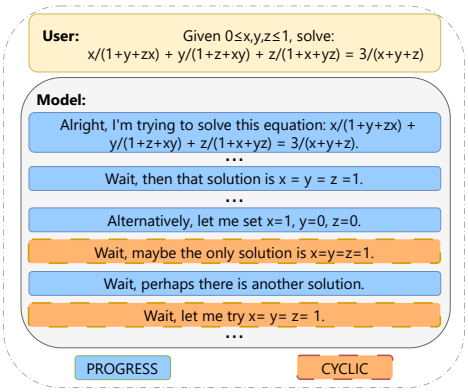


Figure 4: PROGRESS and CYCLIC behaviors illustration.

Teacher-Forced Preferences. To measure whether a model prefers to repeat or to move forward, we compare two candidate next paragraphs at each decision point of a dense trace: repeating the preceding paragraph (CYCLIC) or continuing with the next paragraph that adds new content (PROGRESS). We score the first 32 tokens of each candidate under teacher forcing with the same prefix, and reuse the same decision points and candidates across all pruning checkpoints. We restrict evaluation to decision points that are followed by a PROGRESS paragraph, and define:

$$\Delta_{\text{cyc}} = \sum_{t=1}^{32} \Big[\log p_\theta\big(\mathbf{x}_t^{\text{cyclic}} \mid \text{prefix}, \mathbf{x}_{<t}^{\text{cyclic}}\big) - \log p_\theta\big(\mathbf{x}_t^{\text{progress}} \mid \text{prefix}, \mathbf{x}_{<t}^{\text{progress}}\big) \Big]$$

where a larger Δ_{cyc} indicates a stronger relative preference for CYCLIC over PROGRESS.

As shown in Table 3, pruning progressively shifts the model's preference from progressive toward cyclic continuations at decision points; Appendix D.4 shows a similar shift at randomly sam-